Infrastructure planning

101-0509-10L (ECTS-6KP)

Adaptive Planning

Project Sheet 4 (Weeks 7 and 8)

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0 Purpose and scope of Phase 4

Phase 3 compared fixed infrastructure stages under sampled futures, mainly through Year-1 and Year-40 outcomes. Phase 4 introduces the temporal dimension. It examines how uncertain conditions evolve, when monitoring information becomes available, when decisions are made, and whether infrastructure opens before performance requirements are breached.

The accompanying 04_adaptive_planning.ipynb notebook compares static, predetermined staged and adaptive plans over a 40-year horizon. It reuses the uncertainty definitions and transport surrogate prepared in Notebook 03; it does not retrain the surrogate or repeat the road-assignment design.

The objective is to understand:

- how transient uncertainties are represented as annual trajectories rather than only endpoint values;
- how static, staged and adaptive plans differ in their commitments and timing;
- how monitoring indicators, thresholds, persistence and construction lead times create decision and activation years;
- how the same sampled futures are used to compare plan trajectories and requirement attainment;
- how adaptation tipping points, restoration windows and construction-adjusted decision deadlines are identified;
- whether the adaptive packages open in time and in which order; and
- what is gained and lost by waiting when expenditure and years in failure are kept as separate outcomes.

This phase reports physical performance and the present value of infrastructure expenditure. It does not complete the societal economic appraisal: monetised benefits, lifecycle welfare and residual value are addressed in Notebook 05.

0.1 How to use the Project Sheets

The present exercise sheet provides a short introduction to the course and specific information for the fourth phase of the semester (Weeks 7-8).

The previous (Weeks 1-6) and subsequent phases (Weeks 9–11) are supported by dedicated project sheets.

The project sheets combine conceptual tasks that help students develop the structure of their final report with practical analyses implemented in the accompanying **Jupyter Notebooks (.ipynb)** provided in the GitHub repository.

Throughout the exercise sheets, tasks are grouped into different categories to clarify what is expected from all groups and to provide opportunities for deeper exploration.

We distinguish between:

Supporting Task (optional) Designed to help you understand the uncertainty framework, become familiar with the Jupyter notebooks, and develop an intuition for the rationale behind each analysis. These tasks do not directly produce outputs for the final report. Especially for students with limited experience in probabilistic modelling or Python, we strongly recommend completing them before interpreting the mandatory results.
Mandatory Task Specifying core activities and deliverables that every group is expected to complete and include in the final report.
Advanced Task (optional) Suggesting extensions for groups interested in testing alternative monitoring indicators, trigger thresholds, lead times, uncertainty trajectories, or advanced scenario-discovery methods for adaptive pathways.

Optional refers to the analysis and reporting task. Run the notebook cells in order, including supporting cells required by subsequent analyses. The exercises generally follow this structure: first defining and visualising the uncertainty assumptions, then sampling and exploring their consequences, and finally identifying sensitivity and vulnerability. Consistency checks should be completed before moving from one part to the next.

1 Part 1: Define the planning problem

1.1 Plans, monitoring and performance

A static plan fixes one infrastructure state from the start. A predetermined staged plan commits to package openings in fixed years. An adaptive plan keeps the timing open and activates packages/infrastructural modifications only when monitored evidence satisfies a decision rule.

The table below summarizes some important definitions

Term	Meaning
Monitoring indicator	The observable indicator tracked over time to signal the need for adaptation (e.g., section_combined_daily for counting-section passengers per day).
Trigger threshold	The critical value that the monitoring indicator must cross to initiate an action (the configured monitoring-indicator threshold shown in Section 1.2).
Threshold-crossing year	The first year in which the monitoring indicator crosses the threshold—*before* any persistence requirement is applied.
Decision year	The year in which a formal investment or construction decision is made. Completed observations through the previous year are used to test persistence (see Part 2).
Implementation lead time	The number of years required to plan, procure, and construct the infrastructure after a decision has been made (e.g., environmental approval, tunnelling, or signalling upgrades).
Activation year	Decision year + lead time: the year in which the new infrastructure or operational stage opens for service.
Adaptation tipping point	The first year in which the *current* infrastructure stage can no longer meet the performance target—for example, when the section load reaches or exceeds 110% of capacity, or when the strategic PT share is at or below 35% from Year 20 onward. This is defined precisely in Part 4.
Operational opportunity point	The first year in which adding a package restores all active requirements when the source state fails at least one requirement (Section 4.4). It does not establish an economic benefit–cost verdict.

On top of these, you need to be able to distinguish between four related but different types of events. These events are easy to blur together - keep them separate:

1 **A trigger being reached:** the monitoring indicator has crossed the threshold.
2 **An intervention being decided:** the persistence requirement has been satisfied and a binding decision year has been established.
3 **An intervention becoming operational:** the activation year has arrived and the new infrastructure or service is operational.
4 **The current stage becoming insufficient:** the *transport system itself* fails a critical performance requirement, independently of any decision or trigger. This is the **adaptation tipping point** introduced in Part 4.

A trigger firing and the current stage failing are **not the same event**. A well-designed trigger fires *in advance*, allowing for implementation lead times, so that the intervention becomes operational before the tipping point is reached (see Section 5.4). A poorly designed trigger fires too late.

Tutorial/supporting task

Relate the planning vocabulary to your case study.

- Identify the monitoring indicators, trigger thresholds, persistence requirements and implementation lead times.
- Distinguish threshold-crossing, decision, activation and adaptation-tipping-point years.
- Explain why a trigger is an anticipatory decision rule rather than a performance requirement.

1.2 Analysis settings

This section selects the indicators, requirements and temporary trigger overrides used in this notebook. The default MehrSpur requirements are a strategic PT passenger-kilometre share above 35% from Year 20 onward and a peak section load/capacity below 110% in every year, in order not to provide overcrowded services.

For projects without an active PT counting section, select available corridor-level indicators and requirements that match the project's objectives. Shares should be entered as fractions. Define permanent triggers in code/adaptive_planning.py, and use notebook overrides only for experiments.

Expenditure comparisons use the configured discount rate, which is 2% by default. Investment is booked at opening, operation and maintenance expenditure in each operating year, and the adaptive option premium at time zero. The present value covers expenditure within the analysis horizon; residual value and welfare are assessed in Notebook 05.

Mandatory Task

Define the performance and monitoring settings for your case study.

- Choose physical indicators and state their units, populations and preferred directions.
- Define each requirement, its threshold, comparison operator and first applicable year.
- Choose observable monitoring indicators for both packages and justify why they provide useful warning.
- Record threshold, persistence and lead-time assumptions separately for each package.
- State which changes are temporary notebook overrides and which are permanent plan definitions.

1.3 The six plans

The supplied example contains six plans: baseline, static1, static2, staged1, staged2 and adaptive. They combine two independent packages (stations and tunnel in the MehrSpur example) into four possible operating configurations: neither package, package 1 only, package 2 only, or both packages.

You should now adapt this framework to fit your case study.

Mandatory Task

Specify the plans that will be compared.

- Describe the initial configuration and package schedule or trigger for every plan.
- Check that predetermined opening years, adaptive lead times and package dependencies are internally consistent.
- Explain which decisions remain open in the adaptive plan and which commitments are fixed at the outset.
- Save permanent definitions in PLAN_DEFINITIONS and verify the derived plan table.

1.4 Visual schedules and independent pathways

Schedule plots shown in this section help you understand when each package is available under static and predetermined plans. The illustrative adaptive paths show that either package can open first and that both can be under construction at the same time: they are not ensemble results, and use placeholders to make their functioning visually understandable and comparable to other plans.

Tutorial/supporting task

Use the schedules to verify that the plan definitions produce the intended package states and that no displayed adaptive timing is mistaken for a simulated trigger result.

1.5 Controlled trajectory timing

Adaptive planning uses annual uncertainty trajectories. Two futures may have the same Year-1 and Year-40 values but develop differently, reaching the final value early, gradually, or late. These differences in timing can affect decision and activation years, infrastructure expenditure, and the duration of performance failures.

Here, you compare alternative uncertainty trajectories and examine how their timing affects the adaptive plan.

Tutorial/supporting task

Hold the endpoints and all other inputs fixed while varying one transient uncertainty's shape. Compare threshold crossings, decisions, openings and years in failure; attribute differences only to trajectory timing.

1.6 Detailed fan charts for representative drivers

The notebook illustrates linear, early, late, logistic, almost-flat and random-walk paths. All reach the sampled endpoint, but their intermediate values differ.

Tutorial/supporting task

Inspect the path families for passenger demand and PT preference, or justified alternatives. Confirm common endpoints, plausible intermediate behaviour and the distinction between the illustrative paths and the final ensemble selection.

1.7 Consequences of trajectory timing

The controlled comparison reports requirement failures, package openings and present-value infrastructure expenditure. A later opening may reduce expenditure within the horizon while leaving requirements unprotected for longer.

Tutorial/supporting task

Interpret timing as a trade-off rather than ranking plans by expenditure alone. Separate discounting effects, shorter operating periods and outstanding commitments from physical performance.

2 Part 2: Test the adaptive rules

Part 2 validates the monitoring and trigger logic before the shared-scenarios ensemble. Temporary example overrides apply only to these tests unless they are transferred deliberately to the permanent plan configuration.

2.0 Select monitoring indicators

Candidate monitoring indicators are first observed under the baseline and then compared with future baseline and package performance under the same scenarios. This helps assess whether an indicator provides an early and useful signal of future problems or adaptation needs.

Regression R^2, the regression slope, and Spearman correlation describe how strongly the indicator is associated with later performance and whether the relationship is positive or negative. However, these statistics do not prove that the indicator causes the observed outcome. They also do not automatically determine the best trigger threshold. The final trigger must still be selected and tested by considering persistence, lead times, false or late signals, and the project's planning objectives.

Mandatory Task

Screen and justify a monitoring indicator for each package.

- Compare candidate indicators with later requirement outcomes and with the change created by each package.
- Assess observability, predictive timing and whether the intervention produces a material improvement.
- Interpret the sign and magnitude of the relationship, not R^2 alone.
- Avoid choosing a capacity-related indicator as if it automatically protected a different objective such as minimum PT share.

2.1 Hand-picked trigger examples

Five reproducible scenarios are used to demonstrate how the adaptive decision rules work under different uncertainty trajectories. In each year, the rule uses only observations from completed years, reflecting the information that would actually be available to the decision-maker.

A decision is made only after the monitoring indicator has exceeded its trigger threshold for the required number of consecutive years. The construction lead time is then added to determine when the intervention becomes operational. The two packages are assessed independently, so each package has its own trigger, decision year, and activation year.

Mandatory Task

Validate the trigger rules on the reproducible examples.

- Check the observed indicator, threshold, persistence count, decision year and activation year for both packages.
- Confirm that a decision uses information available through the previous completed year.
- Compare package openings with the applicable performance requirements and explain blank event dates.
- Keep temporary example settings separate from notebook-wide and permanent settings.

2.2 Check a plan under a selected scenario

A detailed single-scenario view combines monitoring, operating configuration, decisions, openings and performance. Both rules trigger on exceedance; using PT share as a monitoring indicator therefore tests rising, not falling, PT share.

Mandatory Task

Use one selected scenario to demonstrate that the plan behaves as specified. Document the rule settings and explain the sequence from observations to decisions, construction, openings and target attainment.

2.3 Compare all plans under one selected scenario

All plans must face the same uncertainty paths. Comparing them under one scenario clarifies how fixed schedules and adaptive decisions alter configurations over time. Present-value expenditure and target attainment remain separate outputs.

Tutorial/supporting task

Compare plan schedules, requirements and expenditure in one shared scenario. Do not infer ensemble robustness from a single example.

2.4 Implementation by uncertainty trajectory shape

The shape heatmap holds sampled endpoints and other inputs fixed while varying two transient trajectories. It reports mean opening year conditional on opening and the share of scenarios without an opening by Year 40.

Tutorial/supporting task

Use the controlled shape comparison to understand timing sensitivity. Report sample counts and keep conditional opening means separate from non-opening frequencies.

2.5 Select trajectories for the remaining analyses

We have now explored how different uncertainty trajectories can affect trigger timing, decisions, and infrastructure activation. In UNCERTAINTY_SHAPES, you can assign a trajectory shape to selected transient uncertainty parameters. Any transient parameter not listed in this setting follows a linear trajectory by default.

The selected shapes are used in all subsequent ensemble analyses, so they can influence the comparison of plans and their performance across scenarios. Each choice should therefore represent a reasonable and clearly documented assumption about how the uncertain parameter may develop over time. It should not be selected after viewing the results simply because it makes a preferred plan perform better.

Mandatory Task

Select and justify the uncertainty trajectory shapes used in Parts 3–5.

- Record every non-linear shape and its rationale.
- Check that the path reaches the sampled Year-40 endpoint and behaves credibly between endpoints.
- Rerun the dependent analyses after changing the selection and retain the final settings with the results.

3 Part 3: Compare plans across scenarios

3.1 Run the shared-scenarios ensemble

The ensemble now evaluates every plan under the same sampled futures and the same selected trajectory shapes. This ensures a consistent comparison: differences in results reflect the plans themselves rather than different uncertainty assumptions.

‘years_in_failure’ counts the number of years in which at least one active performance requirement is not met. Each year is counted only once, even if several requirements fail during that year. It therefore measures the duration of overall system failure, not the total number of individual requirement violations.

Mandatory Task

Create and verify the shared-scenarios plan ensemble.

- Use identical uncertainty draws and trajectory shapes for every plan.
- Retain scenario and plan identifiers, annual outcomes, event dates and present-value infrastructure expenditure.
- Verify the number of scenarios and plans and inspect missing event dates before interpreting results.

3.2 Fan charts by plan family

Fan charts show the median, interquartile range and p10–p90 interval for selected annual outcomes. Requirements appear only over the years in which they apply.

Mandatory Task

Compare how the plan families perform over the full horizon.

- Select indicators that represent the project objectives and important side effects.
- Compare central trajectories and uncertainty bands, not only Year-40 values.
- Identify when plan differences emerge and whether uncertainty bands overlap.
- Interpret section indicators only when a valid counting section is configured.

3.3 Paired Year-40 changes relative to baseline

Each plan is compared with baseline in the same scenario. The empirical cumulative distribution therefore describes paired plan-minus-baseline changes, not differences between unrelated samples or between medians. Modal-share changes are percentage points.

Tutorial/supporting task

Use the paired distributions to identify the direction, spread and consistency of intervention effects. Apply the indicator's preferred direction before labelling a positive or negative difference as beneficial.

3.4 Ensemble spread, central performance and target attainment

The parallel-coordinates plot brings together several dimensions of plan performance, including physical changes, the number of years in which the requirements are met, and the present value of infrastructure expenditure. All displayed axes place favourable performance at the top. The accompanying tables provide the original units, ranges and corresponding top and bottom values.

Each line represents the results of one scenario or an aggregated profile. The median profile summarises the median value on each axis separately. Consequently, it does not necessarily represent a single scenario that actually occurred in the ensemble.

Mandatory Task

Compare typical performance, spread, robustness and expenditure without collapsing them into one score.

- Report the share of scenarios meeting all active requirements throughout their applicable years.
- Compare failure duration as well as the binary robustness result.
- Use physical-unit tables to interpret the normalized parallel-coordinate axes.
- Do not treat minimum expenditure or one preferred indicator as a complete appraisal.

4 Part 4: Diagnose failure and intervention needs

4.1 First failure and duration for each fixed configuration

We now move from evaluating complete plans to identifying when each fixed infrastructure configuration becomes insufficient. An adaptation tipping point is the first year in which a fixed configuration, such as the baseline or a package stage, fails at least one active performance requirement. It is evaluated directly from system performance and independently of whether an adaptive trigger has fired or an intervention has been decided.

The results describe three different aspects of performance. “First failure” shows when problems begin, “failure duration” shows how long they persist, and the “share of scenarios fully meeting all requirements” shows how robust the configuration is across uncertain futures. These measures help identify when adaptation may be needed and whether a package can postpone or avoid failure.

Mandatory Task

Determine adaptation tipping points for every fixed configuration.

- Apply the exact requirement operators and start years defined in Section 1.2.
- Distinguish an initial shortfall, a missed target start date and later deterioration.
- Compare which requirement fails first and how package additions change failure duration.
- Recognise that improved PT attractiveness may increase passenger load and create a competing capacity effect.

4.2 Failure timing and competing requirements

We now examine when failures occur across the ensemble and whether performance later recovers.

The “survival-style curve” shows, for each year, the share of scenarios that have not experienced any failure up to that point. Once a scenario fails, it permanently leaves this curve, even if it meets all requirements again in later years.

The “requirement-specific yearly failure shares” answer a different question: they show the proportion of scenarios failing each requirement in a particular year. These shares can therefore decrease when conditions improve or an infrastructure package restores performance.

The 'endpoint plot' positions each scenario by the Year-40 values of two selected drivers, but colours it by failures over the complete trajectory. It is useful for identifying patterns and possible associations, but it does not show that one variable caused the observed failure or recovery.

Mandatory Task

Explain the timing and mechanism of each requirement failure.

- Compare the same sampled scenarios across fixed configurations.
- Relate weak PT preference to PT-share shortfalls and strong demand to comfort pressure where supported by the results.
- Identify trade-offs when a package improves one requirement while worsening another.
- Avoid interpreting endpoint positions as sufficient descriptions of annual paths.

4.4 Restoration windows and decision deadlines

An “operational opportunity” occurs when the current configuration fails at least one requirement, while adding one feasible package restores all requirements in the same scenario and year.

Consecutive opportunity years form a “restoration window”, indicating when the package must be operational. Subtracting the construction lead time gives the corresponding decision window and latest decision deadline.

Mandatory Task

Identify and interpret restoration windows for every feasible package transition.

- Report the share of scenarios with a window, reachability after monitoring and construction, and the number of restorable years.
- Distinguish the earliest decision needed to protect the full window from the last decision that still provides some restoration.
- Explain blank windows: the source may already comply, or the package may be insufficient.
- Treat the opportunity as technical requirement restoration, not proof of positive net benefits.

4.5 Discover vulnerability mechanisms with PRIM

As in Notebook 3, PRIM can be used to identify combinations of uncertain inputs associated with poor performance. Here, it is applied separately to each active requirement and to both the Baseline and Adaptive trajectories. A trajectory is classified as a failure if the selected requirement is breached in at least one year in which it applies.

PRIM needs enough failed and successful cases to make a meaningful comparison. A box is accepted only if it meets the specified failure-density threshold and narrows at least one uncertain input relative to its full sampled range.

Advanced Task (optional)

Use PRIM to investigate distinct vulnerability mechanisms.

- Choose justified physical Year-40 inputs for the search and report all restricted bounds.
- Report failure frequency and duration together with box density, coverage and mass.
- Keep PT-policy shortfalls and comfort/capacity failures as separate classifications.
- Do not interpret a missing box as proof that the plan has no vulnerability.

4.6 Visual comparison of vulnerability boxes

The two-dimensional rectangles are projections of boxes that may restrict additional inputs. When Adaptive has no qualifying box, a Baseline box may be displayed as a labelled reference with density and coverage recalculated for Adaptive; this is not an Adaptive discovery.

Advanced Task (optional)

Compare the Baseline and Adaptive boxes without hiding higher-dimensional restrictions. Explain differences in failure mechanisms and the limitations created by small, uniform or weakly separated samples.

5 Part 5: Evaluate and improve implementation

5.1 Decisions, construction, openings and non-implementation

Section 5 examines how the adaptive plan is implemented across the sampled futures. It distinguishes between trigger activation, the formal decision, construction, and the year in which a package becomes operational. A triggered decision does not necessarily mean that the package opens within the 40-year analysis horizon, because the activation year also depends on the implementation lead time.

The implementation summary distinguishes four outcomes: the package opens within the horizon; it is committed but scheduled to open after Year 40; it is committed but no opening is recorded; or no decision is made. Any commitment without a recorded opening should be checked carefully. This distinction is important when interpreting package delivery, expenditure, and performance improvements.

Mandatory Task

Quantify implementation outcomes for both adaptive packages.

- Report decision and opening frequencies and their timing distributions.
- Distinguish no decision from a committed package whose activation lies beyond Year 40.
- Compare the operating-state and construction-state trajectories; both packages may be under construction simultaneously.
- Do not interpret implementation probability as requirement attainment.

Tutorial/supporting task

Use the provided implementation summaries and figures to understand how decisions progress into construction and operation.

- Compare the opening-year distributions for both packages, noting that these include only scenarios in which the package opens within the horizon.
- Use the "not yet open" curves to examine all scenarios, including packages that open late or do not open.
- Compare the operating-infrastructure shares with the shares of scenarios in which each package is committed but not yet operational.
- Remember that both packages may be under construction simultaneously, so their construction shares do not need to sum to 100%.

5.2 Opening-time cohorts

For each package, scenarios are grouped by the year in which the package actually opens: early (by Year 15), medium (Years 16–25), late (after Year 25), or no opening. The resulting cohort curves show how performance develops for scenarios with different implementation timings.

However, these cohorts contain different combinations of uncertain conditions. Differences between their curves may therefore reflect both opening timing and the underlying scenarios. They should be interpreted as descriptive comparisons, not as estimates of the causal effect of opening a package earlier.

Tutorial/supporting task

Inspect monitoring and performance trajectories by opening cohort. Keep package-specific cohorts, decisions made within the horizon whose openings occur beyond it and actual openings distinct.

5.4 Did each adaptive package open in time?

Timeliness is assessed through a paired counterfactual comparison within each scenario. For each package, the model removes that package while keeping the other package's observed opening year unchanged; the other package's trigger is not rerun. This identifies when the first requirement failure would occur without the package.

The lead margin is calculated as: first failure year without the package − actual opening year

A positive margin means the package opens before the reference failure, zero means it opens in the same year, and a negative margin means it opens too late. This timing comparison alone does not prove that the package prevents failure, so actual and counterfactual failure durations must also be compared.

Scenarios without an opening or without a counterfactual reference failure are reported as separate categories; missing dates must not be interpreted as Year 0.

Mandatory Task

Evaluate whether each package opened before the relevant counterfactual failure.

- Classify positive, zero and negative lead margins and report initial shortfalls separately.
- Compare actual and counterfactual failure duration; opening before failure does not prove that all requirements were protected.
- Calculate the construction-adjusted decision deadline and compare it with the earliest monitoring-based decision.
- Use restoration windows to explain whether the package could restore full compliance.

5.5 Package order and pathway synthesis

The adaptive plan can follow different implementation pathways across the sampled scenarios: neither package opens, only one package opens, one package opens before the other, or both open in the same year. These pathways describe the sequence of operational stages, not only whether a decision was made.

A no-opening pathway does not necessarily mean that no action was taken. A package may already have been approved and under construction, but its scheduled opening may fall beyond the 40-year analysis horizon.

Mandatory Task

Synthesize the observed adaptive pathways.

- Report the frequency of each opening order using all scenarios, not only those in which both packages open.
- Distinguish operating states from construction commitments.
- Select representative scenarios using coherent simulated timelines rather than mixing median event dates from different runs.
- Connect pathway order to the requirement mechanisms and timeliness results.

5.6 The value and price of waiting

Each scenario compares the adaptive plan with selected predetermined plans under the same uncertain future. This paired comparison shows the trade-off created by waiting: adaptation may reduce the present value of infrastructure expenditure, but it may also increase the number of years in which at least one requirement is not met.

Expenditure savings and additional failure years are reported separately because they measure different outcomes. They must not be combined or interpreted as a monetized welfare or benefit–cost result.

Mandatory Task

Assess the physical and expenditure consequences of waiting.

- Use same-scenario comparisons and state the predetermined comparator plans.
- Explain whether expenditure differences arise from discounting, shorter operation or openings outside the horizon.
- Identify cases where waiting saves expenditure but increases failure duration, and cases where it avoids unnecessary implementation.
- Carry outstanding commitments and the need for economic appraisal forward to Notebook 05.

5.7 Planner-in-the-loop demonstration

The optional planning game places you in the role of the decision-maker. At each checkpoint, you see only the annual observations available up to that point and must decide whether to commit to an

infrastructure package. Decisions become binding in the following year, while packages become operational only after their implementation lead times.

At the end, the scorecard compares your decisions and outcomes with benchmark plans evaluated under the same hidden scenario. This allows you to assess how well your strategy balances timely intervention, requirement attainment, and infrastructure expenditure.

Tutorial/supporting task

Use the game to practise decisions under incomplete information. Compare expenditure, target attainment and failure duration, but do not interpret the scorecard as a full social cost-benefit result.

6 Required outputs for Phase 4

Mandatory Task

By the end of Phase 4, every group should retain:

- a documented set of performance requirements, monitoring indicators, trigger thresholds, persistence rules and lead times;
- verified static, predetermined staged and adaptive plan definitions;
- the selected uncertainty trajectory shapes and a validated shared-scenarios ensemble;
- fan charts and physical-unit summaries comparing plan trajectories, uncertainty spread and robustness;
- fixed-configuration tipping points, failure durations and requirement-specific failure mechanisms;
- restoration windows and construction-adjusted decision deadlines for feasible package transitions;
- implementation frequencies distinguishing decisions, construction, openings and non-implementation;
- package timeliness results based on paired counterfactual pathways;
- pathway-order frequencies and representative construction timelines;
- paired comparisons of expenditure savings and additional years in failure; and
- the completed tables and figures exported to results/04_adaptive_planning/ and figures/04_adaptive_planning/.

A preferred plan based on monetised societal welfare is not required in this phase. Phase 4 establishes whether adaptive rules respond to the selected uncertainties in time, how frequently packages are implemented, and which performance risks remain. Notebook 05 completes the appraisal.

7 Connection to SIA

Phase 4 supports strategic infrastructure planning by linking monitoring, decision timing and implementation to changing system conditions. It extends the uncertainty analysis from fixed endpoint comparisons to adaptive pathways that can respond as information is observed.

In particular, the work supports:

- defining measurable monitoring indicators and transparent decision rules;
- distinguishing policy requirements from anticipatory trigger thresholds;
- testing whether implementation lead times allow interventions to become operational before critical failures;
- comparing fixed commitments with flexible pathways under identical futures;
- identifying restoration opportunities and deadlines for maintaining service objectives;
- documenting residual vulnerabilities and competing performance requirements;
- describing the sequence and timing of infrastructure commitments; and

- preparing the physical and expenditure evidence required for the economic appraisal in Phase 5.

The resulting pathways should be interpreted together with stakeholder knowledge, governance constraints, constructability, monitoring feasibility and effects not represented in the transport model. Adaptive timing is not valuable by definition; its value depends on the decisions it changes, the failures it avoids and the lifecycle consequences assessed in the subsequent appraisal.